Final Project Report
CIS 2100 - Python for Business Analytics
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Group Members and Roles:

Stefani Wilson - Introduction, Dataset Description, Data Cleaning Process, Visualizations

Argel Perez - Exploratory Data Analysis, Business Insights & Interpretation, Limitations, Conclusion

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Introduction

We selected a data set that highlights players performance and statistics. It would be considered to be a part of the sports business domain. This data set would help coaches and managers make informed decisions about trading, playing time per player, and positioning. Our question that we are answering is “How many points does a player contribute for every 10 minutes of play time? ”

Dataset Description

We sourced this data set from Kaggle(see this link). There are 1,628 rows which represent the players and there are 36 columns which represent the stats. In our analysis we used the columns named teams, position, player, mpg_minutes_per_game, ppg_points_per_game. There were issues with missing values and also incorrect data types for categories that led to the data not being able to be analyzed correctly.

Data Cleaning Process

Several cleaning processes were performed to transform the dataset into a format that could be easily analyzed. We ran df_final.info() and df_final.shape to identify that the data set contained 1,628 rows and 36 columns and contained columns with a significant amount of missing data. To fix the issue of this missing data, instead of deleting these columns we changed the NaN value to 0 to help with accurate computations. We cleaned the column names converting them to snake_case and stripping them of leading and trailing whitespaces to ensure the accessing of columns was simple. Several columns such as 3fg%_three-point_field_goal_percentage and ft%_free_throw_percentage were initially recognized as objects. We used pd.to_numeric() with errors= ‘coerce’ to convert them into floats. We also converted the position and team columns into category types. We ran duplicate checks to ensure there wasn’t any redundancy and found that there were not any duplicates in the dataset.

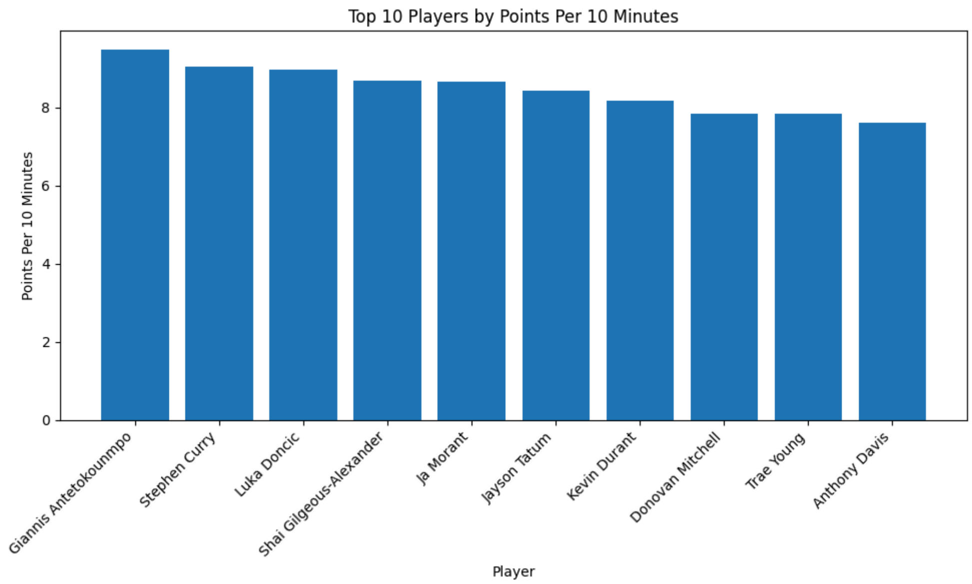
We made these choices to preserve as much data as possible to be able to make accurate assessments and insights about the data.

Exploratory Data Analysis(EDA)

For the exploratory data analysis, we used the cleaned dataset, which had 1,628 rows and 36 columns. The dataset included player information such as name, team, position, games played, minutes per game, and points per game. Our main question was: How many points does a player contribute every 10 minutes of play time? To answer this, we created a new column called points_per_10_minutes by dividing points per game by minutes per game, then multiplying by 10. This helped us compare players more fairly because not all players get the same amount of playing time. After looking at the data, we found that 250 players had valid minutes and points data and out of those players, the average points per 10 minutes was 4.66, and the median was 4.42. This means most players scored around 4 to 5 points for every 10 minutes of playing time. The highest player was Giannis Antetokounmpo, with 9.48 points per 10 minutes. Other top players included Stephen Curry, Luka Doncic, and Ja Morant. We also compared scoring efficiency by position and found that point guards had the highest average at 4.93 points per 10 minutes, while small forwards had the lowest average at 4.35. Overall, the analysis showed that points per game alone does not tell the full story because players can still be efficient scorers even if they play fewer minutes.

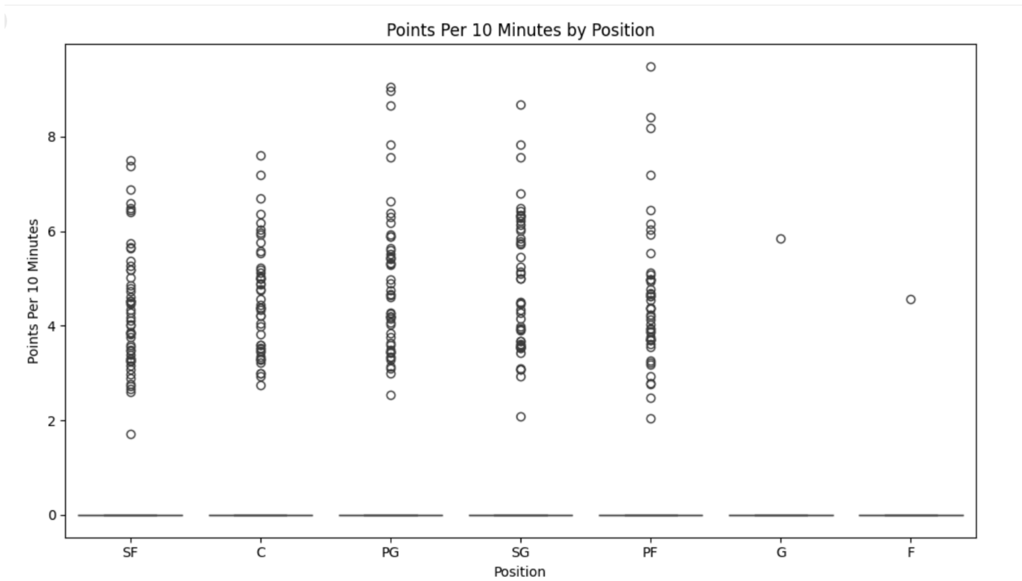
Visualizations

Matplotlib Bar Chart



We chose this chart to showcase the top 10 athletes who have the highest scoring average in a 10 minute timespan. We decided to use the bar chart because it would make it easier for managers to visualize the highest scorers on their team to make informed decisions about who can get more playing time.

SeaBorn Boxplot



We chose to do a boxplot of points per 10 minutes by position. It displays the highest scoring average based on position.

Business Insights & Interpretation

The analysis revealed that points per 10 minutes is a useful way to compare player scoring efficiency. This matters because players do not all play the same amount of minutes per game. Some players may score a lot because they play more, while other players may score efficiently in fewer minutes.This matters to coaches and teams because it can help them make better playing time decisions. If a player scores a lot in limited minutes, the coach may want to give that player more opportunities. The data can be used to help coaches identify players who may be underused or who are producing well when they are on the court. A possible decision from this data would be to review the players with the highest points per 10 minutes and see if they deserve more minutes. Coaches could also compare scoring efficiency by position to understand where the team is strongest or weakest. However, this data should not be the only factor used to make decisions. Coaches should also look at defense, assists, rebounds, turnovers, matchups, and team strategy before changing playing time.

Limitations

One limitation of this project is that missing values were filled with 0. This helped us keep all rows in the dataset, but it may affect the results. Some missing values may mean the stat was not recorded, not that the player actually had 0. Another limitation is that many players had 0 values for minutes and points after cleaning. Because of this, the best scoring-efficiency analysis focused on the 250 players who had valid minutes and points data. This made the results more accurate for the main question. Outliers were also present in the data. Some players had very high points_per_10_minutes compared to the average. These outliers may represent elite players, but they can also affect the overall average. The dataset also does not include everything that matters in basketball. It does not show injuries, defense, matchups, coaching strategy, or player roles. A player's value is not only based on scoring, so points_per_10_minutes should be used as one part of the analysis, not the entire answer. In the future, this analysis could be improved by adding more stats, such as assists, rebounds, turnovers, shooting percentage, and defensive stats. It would also help to filter out players with very low playing time so the results are more reliable.

Conclusion

In conclusion, this project used basketball player statistics to study scoring efficiency. We created a new column called points_per_10_minutes to answer the question: How many points does a player contribute every 10 minutes of play time? The analysis showed that the average player with valid data scored about 4.66 points per 10 minutes. Some players, like Giannis Antetokounmpo, Stephen Curry, and Luka Doncic, scored much higher than average. This shows that some players are more efficient scorers based on their playing time. The business value of this project is that coaches and teams can use this type of analysis to make better decisions. It can help identify players who may deserve more playing time or players who are producing well in limited minutes. A next step for this project would be to include more performance stats, such as assists, rebounds, turnovers, defense, and shooting percentage. This would give a more complete view of each player's overall value.